

Towards Asynchronous Mixed Reality Remote Guidance supported by a Virtual Assistant: Proposal of a Conceptual Model

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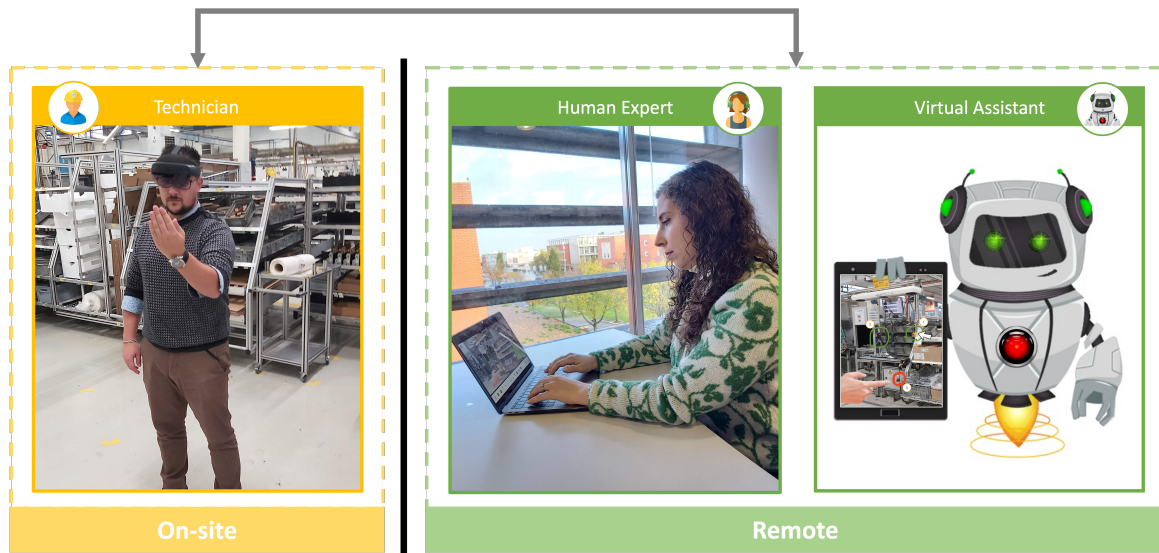


Figure 1: Scenario of asynchronous remote guidance in an Industrial environment, having an on-site technician using Mixed Reality (MR) and being assisted either by a human expert or a Virtual Assistant (VA) to accomplish a given task.

ABSTRACT

Activities of remote guidance are essential in the Industry sector to overcome emerging problems. One important technology to create a common ground between distributed team-members is Mixed Reality (MR). On-site collaborators can receive instructions from remote experts, informing where to act, and what to do. Nevertheless, most studies have focused on synchronous use-cases, where team-members can interact in real-time. Hence, an opportunity exists to explore asynchronous scenarios, in which collaborative actions take place at different times. This position paper proposes a vision for combining MR and a Virtual Assistant (VA) for supporting asynchronous remote guidance in Industrial Scenarios. This combination can be highly flexible in supporting most simpler tasks, taking advantage of existing instructions from previous sessions. Also important, a VA can be constantly available, reducing time-to-response, while supporting various individuals with distinct needs. These and other arguments in favor of this position are presented and future directions are proposed.

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1 INTRODUCTION

Scenarios of remote collaboration are a promising area of research for Mixed Reality (MR) technologies [3, 13, 17, 24, 26] which can be applied to numerous use-cases associated with Industry 4.0, a field with an increasing level of adaptability towards the growing demands of mass customization [1, 4, 5, 19].

Solutions exploring MR allow interconnection of physically separated team-members, trying to achieve a common goal through unique capabilities, thus, creating a shared understanding for situation understanding, improved communication, and generation of augmented content, which can be overlapped on top of the real-world context, thereby merging the real and virtual worlds [2, 8, 23, 26]. By doing so, remote experts can generate situated instructions, allowing an on-site member to understand how to handle a given problem (where to act, and what to do), e.g., which may be causing an assembly line to be stopped, resulting in a serious impact on a given factory [4, 5, 18, 19].

Thus far, the vast majority of the research efforts on remote collaboration supported by MR have been devoted to the study of synchronous situations, i.e., in which all team-members can interact and communicate simultaneously in real-time, being able to repeat lost or ambiguous information through different modalities (e.g., speech, or non-verbal, etc.). Nevertheless, due to various constraints

(e.g., distinct time zones, etc.), team members may not always be available to participate in this type of knowledge exchange. Under these circumstances, remote asynchronous collaboration must be considered, allowing teams to work collectively at different moments in time towards a shared end result. This activity requires information retention, and its consumption at a later time [9, 15, 16].

Therefore, an opportunity exist to explore asynchronous use-cases into a number of potential directions, where collaborative actions of distributed team-members take place over an extended time duration. As a matter of fact, industrial settings offer a compelling and richer environment where to conduct this research, given the fact that they usually incorporate distributed locations working towards the creation of the distinct components of a final product. Also relevant, they also have various shifts with a multidisciplinary workforce, divided in different levels of expertise. In this vein, information sharing is consistently required by stakeholders at different times [4, 5, 9, 19].

Another relevant field of study associated with Industry 4.0 is Artificial intelligence (AI), a field dedicated to solve intelligence [6, 7, 11]. To contextualize, the use of an intelligent Virtual Assistant (VA) has evolved immensely over the last decades, focusing on improving human-like qualities of VAs like multimodality, believability, and naturalness as an interlocutor. Despite this, a very reduced number of works have focused on aspects of human-virtual assistant collaboration [10, 20, 21, 23, 25, 27]. Although MR and AI are distinct technologies, both have experienced unprecedented growth in industry use-cases and can be utilized together for interactive experiences.

To elaborate, there is an enormous potential for integrating AI models since recent algorithms have achieved great success extracting data from images and videos. In turn, this data can be processed and used to create 2D and 3D instructions that can be consumed in an MR setting [1, 23]. If all this is combined with the possibility of interacting with an intelligent VA (Figure 1), it may be possible to properly assist on-site members requiring additional information to overcome a concrete situation. As such, the authors strongly believe these fields can be the next prominent direction in upcoming years, in particular for remote asynchronous scenarios, increasing autonomy, adaptability/flexibility, and effectiveness when a remote counterpart is not available at a given moment.

In this vein, this position paper describes a conceptual model for supporting asynchronous remote guidance in industrial scenarios through the use of MR and a Virtual Assistant (VA). The arguments in favor of this proposal are described, as well as future directions. To be clear, the purpose of introducing MR and VAs in this context is to facilitate how on-site members have access to distributed knowledge for handling specific problems, rather than to completely replace human involvement.

2 ARGUMENTS FOR OUR PROPOSAL: VIRTUAL ASSISTANTS AND MR FOR ASYNCHRONOUS REMOTE GUIDANCE

Next, various arguments to support our position are described.

- Industry 4.0 moves towards combining multiple fields of expertise to improve workers' productivity, as is the case of MR, AI, Big data, Cloud computing, and others;
- Intelligent VAs have become mature enough to support more challenging scenarios as is the case of asynchronous remote guidance;
- For some simpler scenarios, the lack of the human element may be overcome thanks to VA capacity to analyze, process, and generate instructions with a rather similar efficiency;
- Solutions using MR allow sharing of contextual data for situation understanding, as well as the consumption of AI-generated

instructions and their superimposition over the real-world environment;

- Requesting assistance from a VA can be made under many circumstances:
 - Handling multiple on-site individuals at once, even with distinct tasks;
 - Quickly identifying similar or identical tasks, reducing time-to-response;
 - Avoiding the need to recall remote Human experts - allowing them to focus on more demanding problems;
 - Re-use previous instructions based on the knowhow obtained from previous remote sessions;
 - Using each remote session to train its algorithms, thus improving instructions quality while increasing the database size and diversity;
 - Similar methods may be used to handle different application scenarios, not being limited to industrial use-cases.

3 CONCEPTUAL MODEL FOR HUMAN-VIRTUAL ASSISTANT ASYNCHRONOUS MR REMOTE GUIDANCE

The conceptual model described in the present section (Figure 2) suggests distinct modules for industrial MR-remote guidance between an on-site member and a remote virtual assistant. The boxes represent different key elements with arrows loosely indicating a flow between elements. To create a novel approach, a conceptualization taking inspiration from the Human-to-Human collaboration model of Marques et al. [12, 13] was conducted. In particular, the new model uses the on-site member representation, followed by a proposal considering the role of a remote VA which replaces the typical remote human expert. Also, there was no particular concern with specific supporting technologies, but mostly with the steps required to accomplish the intended task.

The on-site member begins the interaction process by collecting contextual data from the surrounding environment, which will be considered accordingly throughout the interaction pipeline resulting in the creation of multimodal MR content. First, it feeds the registration data module to enable the identification of regions of interest. The capture content module provides the grounds for spatially (and temporally) augmentation aligned with the physical world after the remote VA intervenes. To elaborate, this data (e.g., a video feed, a virtual scene, tangible landmarks, or a reproduction of the physical context) can be used to promote interaction and communication with the assistant, creating a representation of the current problem that needs to be addressed.

In turn, the interaction bot will use this information, as well as multimodal communication cues to analyze the task status, resulting in one of three situations: 1) If the problem is not yet overcome, the shared context must be analyzed to determine how a solution could be put together. This will be done by comparing the information received against the data stored in a knowledge database. This way, if the problem at hand has occurred before in previous sessions, and if an existing solution already exists, it may be re-used to guide the on-site member. Otherwise, similar situations may be examined, as well as manuals, videos, reports, or other relevant sources of knowledge to generate novel instructions capable of addressing the problem needing a solution. The new set of instructions (composed by assessing an asset database) will be used as a new entrance into the knowledge database, extending the range and capacity of the VA to address problems moving forward. Likewise, the instructions produced will also advance to the compositing module. Subsequently, a set of MR content is obtained, which the on-site member can experience through the output presentation module. Following the VA analysis, the output may be provided using different modalities

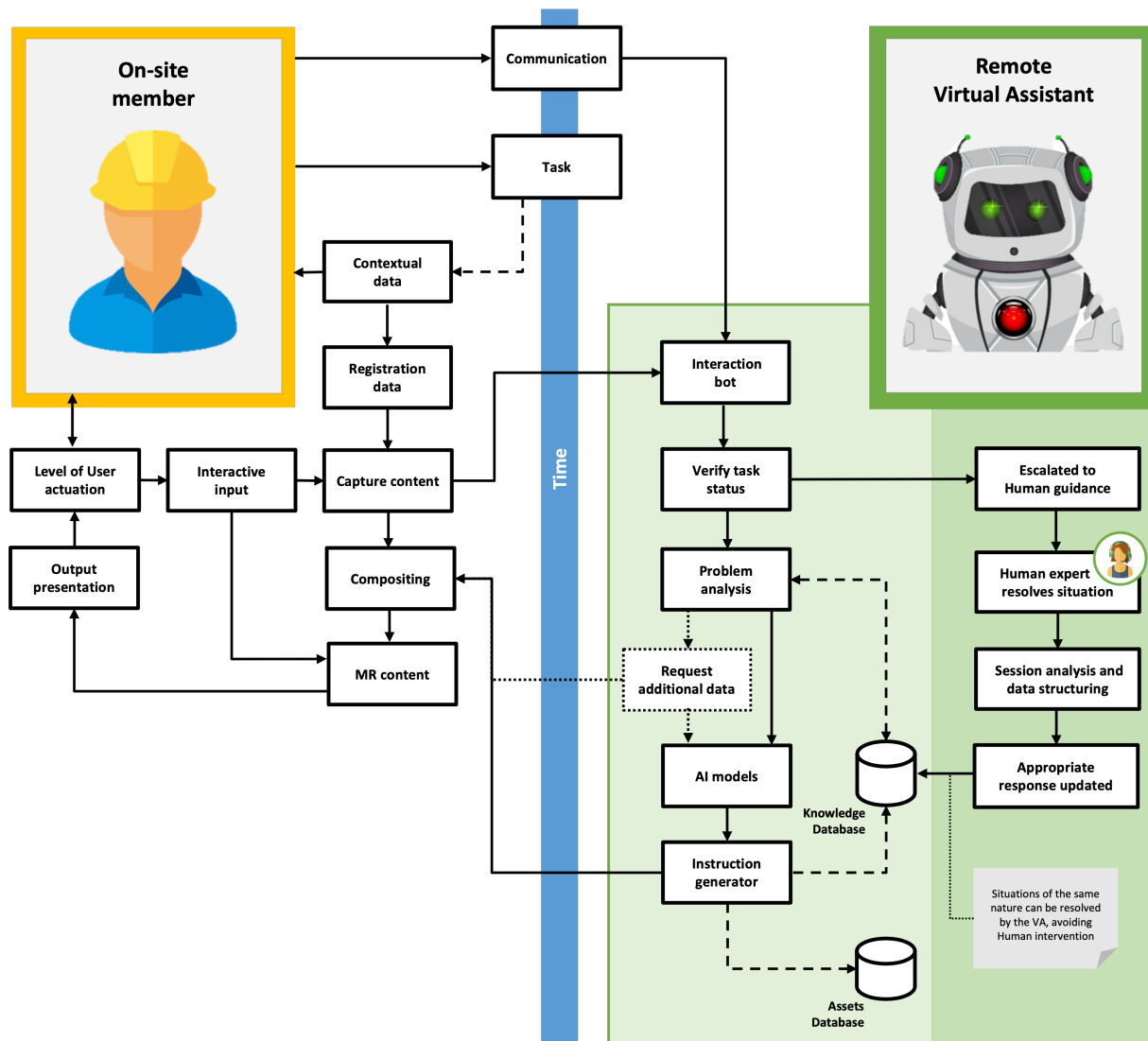


Figure 2: Conceptual Model illustrating the necessary elements for a scenario of asynchronous remote guidance between an on-site member (left) and a remote virtual assistant (right). Inspired by the Human-to-Human collaboration model of Marques et al. [12, 13].

based on the variables considered (e.g., environment context, on-site member characteristics, task difficulty, type of instruction generated, etc.). Allowing the on-site member to express himself after analyzing the content received, the level of user actuation module offers the capacity to alter how the scene is augmented through the compositing module, i.e., selecting which elements should be visible, exploring certain aspects of the virtual scene, or using the input module for interacting with some attributes of the shared content. In addition, the VA may request additional information in some cases, in order to generate instructions with a higher level of confidence when a certain level is not attained. In other words, collecting additional data for gaining higher knowhow regarding the problem being addressed. This will also help to configure how the multimodal output is presented.

Moreover, (2) if the problem persists and the VA does not have the capacity to guide the on-site member, the process is assigned to a third party, i.e., a remote human expert to resolve the situation. To complement this scenario, an analysis of the collaborative session

between the on-site and remote team-members can occur, allowing the generation of a structured response that can be used to update the knowledge database. Thus, in the future, situations of the same nature may be resolved by the VA, avoiding Human intervention;

Last, (3) if the problem is resolved, the interaction session is complete and the connection is terminated, with the VA remaining in a state of readiness for further guidance sessions, if necessary.

As a final remark, the terminology used for each of the modules used was harmonized, as best as possible, taking into account nomenclature already used in recent literature [2, 8, 14, 17, 22, 24, 28–30].

4 FUTURE DIRECTIONS: A PROPOSAL

Many issues need to be addressed by the research community with the goal of exploring the impact of using a VA for supporting MR asynchronous remote guidance, in particular:

- **Design and development of frameworks** - creation of multi-platform MR tools, and VAs integrating suitable AI models;

- **Explore multimodality features** - supporting the use of distinct modalities either as input to explain a given problem, or as output to explain how to conduct a set of instructions;
- **Creation of data-bases** - establishment of the necessary data-bases, as well as ensuring their maintenance over time. Plus, guarantee the databases are populated with relevant content to be used by the VA analysis module;
- **Definition of modular architectures** - ensure that a scalable and generic approach is used, which may be applied to various use-cases;
- **Solution evaluation** - apply distinct evaluation methods to appraise the VA capacity to address problems with distinct levels of complexity, e.g., through performance assessment, as well as user evaluation in various levels, starting with laboratory tests and moving into industrial scenarios down the line;
- **Iteratively move into more complex scenarios** - start addressing tasks with similar problems as existing records in the knowledge database and progressively move into more challenging multimodal situations requiring AI models to analyze and propose unexplored scenarios;
- **Explore different data collection methods** - provide a larger data-set for the VA to analyze and build better instructions from, capitalizing on various kinds of sensory information from the real world;
- **Enable the VA to have multiple portraits** - allow personalization according to the user profile, individualizing how the instructions are generated and provided;
- **Mimic Remote experts characteristics** - provide MR instructions with a level of realism similar to existing human experts, to better support on-site members;
- **Capitalize into MR technological capabilities - Avatar generation** - explore its capacity to improve perception, awareness and empty, e.g., use avatars to provide a representation of the VA, which the user can interact directly with and more easily relate to while collaborating with this entity.

5 CONCLUDING REMARKS

In summary, the authors strongly believe that an opportunity exists to explore asynchronous remote use-cases in Industrial environments. In particular, to verify the possibility of combining VA and MR for supporting on-site team-members during guidance scenarios.

Overall, several topics remain unanswered, illustrating that extensive research in multiple disciplines is still necessary. To elaborate, future work must seek to determine the extent to which VA and MR can be used for properly guiding Human individuals, starting by asserting if there are observable differences in the use of distinct interaction and MR generation methods during asynchronous remote scenarios. In particular, to verify if on-site individuals can tell the difference between being assisted by remote human experts and a virtual assistant. Also, how this research can address synchronous scenarios down the line. Another relevant aspect is to analyze how existing surveys and taxonomies include the VA entity in the collaborative process, and, if that is not the case, propose extended versions of existing approaches that help advance the state of the art, while generating a better understanding of how collaboration can be perceived by the research community.

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